

2.3 The Formal Viewpoint

Sometimes it is good to step back from the forest of details and look for general patterns. In this rather brief section we will first describe the general pattern of homology by axioms, then we will look at some common formal features shared by many of the constructions we have made, using the language of categories and functors which has become common in much of modern mathematics.

Axioms for Homology

For simplicity let us restrict attention to CW complexes and focus on reduced homology to avoid mentioning relative homology. A (reduced) **homology theory** assigns to each nonempty CW complex X a sequence of abelian groups $\tilde{h}_n(X)$ and to each map $f: X \rightarrow Y$ between CW complexes a sequence of homomorphisms $f_*: \tilde{h}_n(X) \rightarrow \tilde{h}_n(Y)$ such that $(fg)_* = f_*g_*$ and $\mathbb{1}_* = \mathbb{1}$, and so that the following three axioms are satisfied.

- (1) If $f \simeq g: X \rightarrow Y$, then $f_* = g_*: \tilde{h}_n(X) \rightarrow \tilde{h}_n(Y)$.
- (2) There are boundary homomorphisms $\partial: \tilde{h}_n(X/A) \rightarrow \tilde{h}_{n-1}(A)$ defined for each CW pair (X, A) , fitting into an exact sequence

$$\cdots \xrightarrow{\partial} \tilde{h}_n(A) \xrightarrow{i_*} \tilde{h}_n(X) \xrightarrow{q_*} \tilde{h}_n(X/A) \xrightarrow{\partial} \tilde{h}_{n-1}(A) \xrightarrow{i_*} \cdots$$

where i is the inclusion and q is the quotient map. Furthermore the boundary maps are natural: For $f: (X, A) \rightarrow (Y, B)$ inducing a quotient map $\bar{f}: X/A \rightarrow Y/B$, there are commutative diagrams

$$\begin{array}{ccc} \tilde{h}_n(X/A) & \xrightarrow{\partial} & \tilde{h}_{n-1}(A) \\ \downarrow \bar{f}_* & & \downarrow f_* \\ \tilde{h}_n(Y/B) & \xrightarrow{\partial} & \tilde{h}_{n-1}(B) \end{array}$$

- (3) For a wedge sum $X = \bigvee_{\alpha} X_{\alpha}$ with inclusions $i_{\alpha}: X_{\alpha} \hookrightarrow X$, the direct sum map $\bigoplus_{\alpha} i_{\alpha*}: \bigoplus_{\alpha} \tilde{h}_n(X_{\alpha}) \rightarrow \tilde{h}_n(X)$ is an isomorphism for each n .

Negative values for the subscripts n are permitted. Ordinary singular homology is zero in negative dimensions by definition, but interesting homology theories with nontrivial groups in negative dimensions do exist.

The third axiom may seem less substantial than the first two, and indeed for finite wedge sums it can be deduced from the first two axioms, though not in general for infinite wedge sums, as an example in the Exercises shows.

It is also possible, and not much more difficult, to give axioms for unreduced homology theories. One supposes one has relative groups $h_n(X, A)$ defined, specializing to absolute groups by setting $h_n(X) = h_n(X, \emptyset)$. Axiom (1) is replaced by its

obvious relative form, and axiom (2) is broken into two parts, the first hypothesizing a long exact sequence involving these relative groups, with natural boundary maps, the second stating some version of excision, for example $h_n(X, A) \approx h_n(X/A, A/A)$ if one is dealing with CW pairs. In axiom (3) the wedge sum is replaced by disjoint union.

These axioms for unreduced homology are essentially the same as those originally laid out in the highly influential book [Eilenberg & Steenrod 1952], except that axiom (3) was omitted since the focus there was on finite complexes, and there was another axiom specifying that the groups $h_n(\text{point})$ are zero for $n \neq 0$, as is true for singular homology. This axiom was called the ‘dimension axiom,’ presumably because it specifies that a point has nontrivial homology only in dimension zero. It can be regarded as a normalization axiom, since one can trivially define a homology theory where it fails by setting $h_n(X, A) = H_{n+k}(X, A)$ for a fixed nonzero integer k . At the time there were no interesting homology theories known for which the dimension axiom did not hold, but soon thereafter topologists began studying a homology theory called ‘bordism’ having the property that the bordism groups of a point are nonzero in infinitely many dimensions. Axiom (3) seems to have appeared first in [Milnor 1962].

Reduced and unreduced homology theories are essentially equivalent. From an unreduced theory h one gets a reduced theory $\tilde{h}$ by setting $\tilde{h}_n(X)$ equal to the kernel of the canonical map $h_n(X) \rightarrow h_n(\text{point})$. In the other direction, one sets $h_n(X) = \tilde{h}_n(X_+)$ where X_+ is the disjoint union of X with a point. We leave it as an exercise to show that these two transformations between reduced and unreduced homology are inverses of each other. Just as with ordinary homology, one has $h_n(X) \approx \tilde{h}_n(X) \oplus h_n(x_0)$ for any point $x_0 \in X$, since the long exact sequence of the pair (X, x_0) splits via the retraction of X onto x_0 . Note that $\tilde{h}_n(x_0) = 0$ for all n , as can be seen by looking at the long exact sequence of reduced homology groups of the pair (x_0, x_0) .

The groups $h_n(x_0) \approx \tilde{h}_n(S^0)$ are called the **coefficients** of the homology theories h and $\tilde{h}$, by analogy with the case of singular homology with coefficients. One can trivially realize any sequence of abelian groups G_i as the coefficient groups of a homology theory by setting $h_n(X, A) = \bigoplus_i H_{n-i}(X, A; G_i)$.

In general, homology theories are not uniquely determined by their coefficient groups, but this is true for singular homology: If h is a homology theory defined for CW pairs, whose coefficient groups $h_n(x_0)$ are zero for $n \neq 0$, then there are natural isomorphisms $h_n(X, A) \approx H_n(X, A; G)$ for all CW pairs (X, A) and all n , where $G = h_0(x_0)$. This will be proved in Theorem 4.59.

We have seen how Mayer-Vietoris sequences can be quite useful for singular homology, and in fact every homology theory has Mayer-Vietoris sequences, at least for CW complexes. These can be obtained directly from the axioms in the follow-

ing way. For a CW complex $X = A \cup B$ with A and B subcomplexes, the inclusion $(B, A \cap B) \hookrightarrow (X, A)$ induces a commutative diagram of exact sequences

$$\begin{array}{ccccccccc} \cdots & \longrightarrow & h_{n+1}(B, A \cap B) & \longrightarrow & h_n(A \cap B) & \longrightarrow & h_n(B) & \longrightarrow & h_n(B, A \cap B) & \longrightarrow & \cdots \\ & & \downarrow \approx & & \downarrow & & \downarrow & & \downarrow \approx & & \\ \cdots & \longrightarrow & h_{n+1}(X, A) & \longrightarrow & h_n(A) & \longrightarrow & h_n(X) & \longrightarrow & h_n(X, A) & \longrightarrow & \cdots \end{array}$$

The vertical maps between relative groups are isomorphisms since $B/(A \cap B) = X/A$. Then it is a purely algebraic fact, whose proof is Exercise 38 at the end of the previous section, that a diagram such as this with every third vertical map an isomorphism gives rise to a long exact sequence involving the remaining nonisomorphic terms. In the present case this takes the form of a Mayer-Vietoris sequence

$$\cdots \longrightarrow h_n(A \cap B) \xrightarrow{\varphi} h_n(A) \oplus h_n(B) \xrightarrow{\psi} h_n(X) \xrightarrow{\partial} h_{n-1}(A \cap B) \longrightarrow \cdots$$

Categories and Functors

Formally, singular homology can be regarded as a sequence of functions H_n that assign to each space X an abelian group $H_n(X)$ and to each map $f: X \rightarrow Y$ a homomorphism $H_n(f) = f_*: H_n(X) \rightarrow H_n(Y)$, and similarly for relative homology groups. This sort of situation arises quite often, and not just in algebraic topology, so it is useful to introduce some general terminology for it. Roughly speaking, ‘functions’ like H_n are called ‘functors,’ and the domains and ranges of these functors are called ‘categories.’ Thus for H_n the domain category consists of topological spaces and continuous maps, or in the relative case, pairs of spaces and continuous maps of pairs, and the range category consists of abelian groups and homomorphisms. A key point is that one is interested not only in the objects in the category, for example spaces or groups, but also in the maps, or ‘morphisms,’ between these objects.

Now for the precise definitions. A **category** $\mathcal{C}$ consists of three things:

- (1) A collection $\text{Ob}(\mathcal{C})$ of **objects**.
- (2) Sets $\text{Mor}(X, Y)$ of **morphisms** for each pair $X, Y \in \text{Ob}(\mathcal{C})$, including a distinguished ‘identity’ morphism $\mathbb{1} = \mathbb{1}_X \in \text{Mor}(X, X)$ for each X .
- (3) A ‘composition of morphisms’ function $\circ: \text{Mor}(X, Y) \times \text{Mor}(Y, Z) \rightarrow \text{Mor}(X, Z)$ for each triple $X, Y, Z \in \text{Ob}(\mathcal{C})$, satisfying $f \circ \mathbb{1} = f$, $\mathbb{1} \circ f = f$, and $(f \circ g) \circ h = f \circ (g \circ h)$.

There are plenty of obvious examples, such as:

- The category of topological spaces, with continuous maps as the morphisms. Or we could restrict to special classes of spaces such as CW complexes, keeping continuous maps as the morphisms. We could also restrict the morphisms, for example to homeomorphisms.
- The category of groups, with homomorphisms as morphisms. Or the subcategory of abelian groups, again with homomorphisms as the morphisms. Generalizing